

```
4ITA2@debian:~/ARPIT_240911630/Lab8$ gcc question1.c -o q1
```

```
4ITA2@debian:~/ARPIT_240911630/Lab8$ ./q1
```

```
Enter Allocation matrix (5x3):
```

```
0 1 0
```

```
2 0 0
```

```
3 0 2
```

```
2 1 1
```

```
0 0 2
```

```
Enter Max matrix (5x3):
```

```
7 5 3
```

```
3 2 2
```

```
9 0 2
```

```
2 2 2
```

```
4 3 3
```

```
Enter Available vector (3 values): 3 3 2
```

```
Allocation:
```

```
A B C
```

```
P0 -> 0 1 0
```

```
P1 -> 2 0 0
```

```
P2 -> 3 0 2
```

```
P3 -> 2 1 1
```

```
P4 -> 0 0 2
```

```
Max:
```

```
A B C
```

```
P0 -> 7 5 3
```

```
P1 -> 3 2 2
```

```
P2 -> 9 0 2
```

```
P3 -> 2 2 2
```

```
P4 -> 4 3 3
```

```
Need:
```

```
A B C
```

```
P0 -> 7 4 3
```

```
P1 -> 1 2 2
```

```
P2 -> 6 0 0
```

```
P3 -> 0 1 1
```

```
P4 -> 4 3 1
```

```
Available: 3 3 2
```

```
System is in SAFE state.
```

```
Safe sequence: P1 -> P3 -> P4 -> P0 -> P2
```

```
Testing request from P1: (1,0,2)
```

```
Request GRANTED.
```

Request GRANTED.

Updated Allocation:

	A	B	C
P0 ->	0	1	0
P1 ->	3	0	2
P2 ->	3	0	2
P3 ->	2	1	1
P4 ->	0	0	2

Updated Need:

	A	B	C
P0 ->	7	4	3
P1 ->	0	2	0
P2 ->	6	0	0
P3 ->	0	1	1
P4 ->	4	3	1

Updated Available: 2 3 0

Safe sequence: P1 -> P3 -> P4 -> P0 -> P2

Testing request from P4: (3,3,0)

Request CANNOT be granted immediately.

Testing request from P0: (0,2,0)

Request GRANTED.

Updated Allocation:

	A	B	C
P0 ->	0	3	0
P1 ->	2	0	0
P2 ->	3	0	2
P3 ->	2	1	1
P4 ->	0	0	2

Updated Need:

	A	B	C
P0 ->	7	2	3
P1 ->	1	2	2
P2 ->	6	0	0
P3 ->	0	1	1
P4 ->	4	3	1

Updated Available: 3 1 2

Safe sequence: P3 -> P1 -> P2 -> P0 -> P4

4ITA2@debian:~/ARPIT_240911630/Lab8\$ \

```
• 4ITA2@debian:~/ARPIT_240911630/Lab8$ gcc question2.c -o q2
• 4ITA2@debian:~/ARPIT_240911630/Lab8$ ./q2
Enter Allocation matrix (5x3):
0 1 0
2 0 0
3 0 3
2 1 1
0 0 2
Enter Request matrix (5x3):
0 0 0
2 0 2
0 0 0
1 0 0
0 0 2
Enter Available vector (3 values): 0 0 0
```

Allocation:

	A	B	C
P0 ->	0	1	0
P1 ->	2	0	0
P2 ->	3	0	3
P3 ->	2	1	1
P4 ->	0	0	2

Request:

	A	B	C
P0 ->	0	0	0
P1 ->	2	0	2
P2 ->	0	0	0
P3 ->	1	0	0
P4 ->	0	0	2

No deadlock detected.

Completion order: P0 -> P2 -> P3 -> P4 -> P1

Testing part (b): P2 requests one additional C.

Deadlock occurs after extra request.

Deadlocked processes: P1 P2 P3 P4